



(Company No.: 198301005860)

الجامعة الإسلامية العالمية ماليزيا
INTERNATIONAL ISLAMIC UNIVERSITY MALAYSIA
يُونُوسِيَّتِي اِسْلَامُ اَنْتَا اِبْغْيَا مِلْدِيَا

Garden of Knowledge and Virtue

MECHATRONICS SYSTEM INTEGRATION : MCTA 3203

LABORATORY REPORT

SECTION 1

TITLE : Controlling a DC Motor Using L298P Motor Driver Shield and GPIO (WEEK 5)

LECTURER'S NAME: ASSOC. PROF. IR. TS. DR. ZULKIFLI BIN ZAINAL ABIDIN

DATE OF EXPERIMENT : 5 NOVEMBER 2025

DATE OF SUBMISSION : 12 NOVEMBER 2025

GROUP : 4

NO.	NAME	MATRIC NO.
1	MOHAMMAD FARISH ISKANDAR BIN MOHD SYAHIDIN	2311779
2	AHMAD HARITH IMRAN BIN MOHD YUSOF	2311581
3	ARIFA AQILAH BINTI ABDUL HALIM	2311530

ACKNOWLEDGEMENTS

We would like to express our heartfelt appreciation to Dr. Zulkifli Bin Zainal Abidin for his continuous guidance, support, and insightful explanations throughout this experiment. His expertise greatly helped us understand the concepts of interfacing and digital control using Arduino. We would also like to thank our lab assistant for providing technical assistance and ensuring that the experiment ran smoothly. Finally, we extend our gratitude to our classmates for their cooperation and teamwork during the lab session.

ABSTRACT

This experiment aims to understand how the L298P Motor Driver Shield works in controlling a DC motor using an Arduino. The main objectives are to control the motor's speed and direction by using GPIO pins and Pulse Width Modulation (PWM). The circuit was built by connecting the L298P shield to the Arduino and wiring the DC motor to the motor terminals. The Arduino code was uploaded to move the motor forward and backward while adjusting the PWM value to change its speed. The results showed that higher PWM values produced faster motor rotation, while lower values made it slower. The experiment successfully demonstrated that PWM can effectively control motor speed and that the L298P driver provides an easy and safe way to manage motor direction and power.

TABLE OF CONTENT

ABSTRACT	3
TABLE OF CONTENT	4
1.0 INTRODUCTION	5
2.0 MATERIAL AND EQUIPMENT	6
3.0 EXPERIMENTAL SETUP	7
4.0 METHODOLOGY	8
5.0 DATA COLLECTION	12
6.0 DATA ANALYSIS	12
7.0 RESULTS	14
8.0 DISCUSSION	17
9.0 CONCLUSION	19
10.0 RECOMMENDATIONS	20
11.0 REFERENCES	21
12.0 APPENDICES	22
13.0 STUDENT’S DECLARATION	24

1.0 INTRODUCTION

This experiment focuses on learning how to control a DC motor using the L298P Motor Driver Shield. The main goal is to understand how to use the L298P H-Bridge chip to operate a DC motor. Specifically, the objectives include learning how the shield functions, using digital pins to control the motor's direction, and applying Pulse Width Modulation (PWM) to accurately adjust the motor's speed.

The L298P shield is designed around the L298P H-Bridge chip, which can drive two DC motors simultaneously. Since DC motors typically require a higher voltage and current (around 9V–12V) than the Arduino's 5V output, the shield uses an external power supply to operate the motors while receiving control signals from the Arduino. The motor's direction is controlled through the digital pins IN1 and IN2 (D12 and D13 for Motor A), while the speed is managed using PWM via the ENA pin (D3). PWM works by rapidly switching the motor's power on and off. By adjusting the duty cycle, which represents the percentage of time the signal remains on, the average voltage sent to the motor changes and thus affects its speed.

Based on this principle, it is expected that changing the logic levels of the direction pins will cause the motor to rotate forward or backward. The hypothesis is that the motor's speed will increase proportionally with higher PWM values. For example, a PWM value of 255 should make the motor spin at maximum speed, while a lower value such as 64 should make it rotate more slowly. The experiment will test these variations to observe how PWM affects motor performance.

2.0 MATERIAL AND EQUIPMENT

1. Arduino UNO
2. L298P Motor Driver Shield
3. 2 DC Motor (6V–12V)
4. External Power Supply (9V/12V)
5. Jumper Wires

3.0 EXPERIMENTAL SETUP

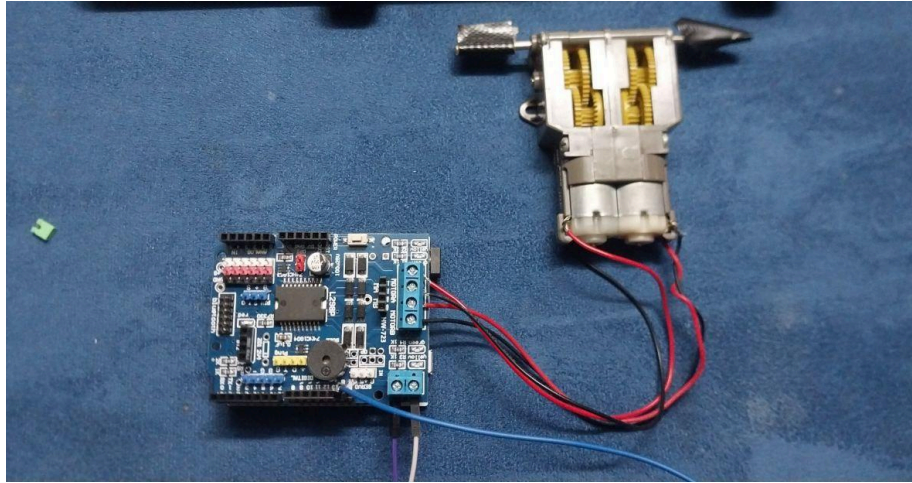


Figure A: 2 DC motor connected to L298P motor driver

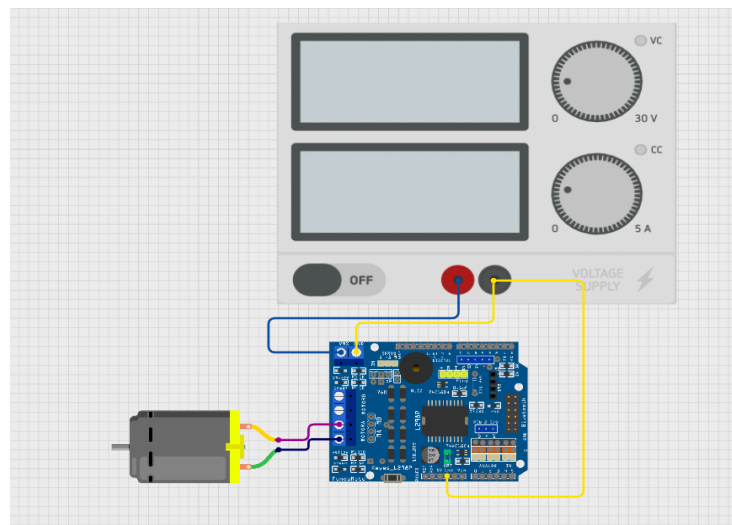


Figure B: DC motor connected to L298P in Cirkuit Designer

This Motor Driver Shield based on L298P motor driver IC is capable of driving two brushed motors, with speed, direction, start and stop simultaneously. The setup for this experiment that drives two DC motors simultaneously using an external power supply that inputs constant 6.5V throughout the whole experiment. The DC motor was connected to the pin provided which was Motor A and Motor B on the motor driver. The VMS and GND port on the motor driver was connected directly to the external power supply with the GND pin connecting the arduino and

motor driver was connected to the ground port of the external power supply. The OPT jumper on the motor driver must be removed to make sure that the DC motor runs directly from the external power supply and not the arduino as there was insufficient current and the inductive voltage spike can destroy the chip of the microcontroller.

4.0 METHODOLOGY

1. The circuit was assembled according to the given circuit diagram and connection instructions.
2. The rotation of both motors was observed in the forward and reverse directions, ensuring the control logic was correct.
3. The PWM values for ENA and ENB in the analogWrite commands were modified using 255, 128, 64 to observe how changes in the duty cycle affected the speeds of Motor A and Motor B independently and simultaneously.
4. Observations were documented by estimating the speed for each motor under various PWM and direction conditions to fulfill the task of dual-motor control.

Circuit Assembly

- The L298P Motor Driver Shield was mounted directly on top of the Arduino UNO board.
- Motor A and Motor B were connected to their respective terminals on the motor driver shield.
- An external power supply (9V/12V) was connected to the shield's power input terminals (VMS and GND).
- The ground (GND) connection was shared between the motor driver shield and the Arduino to ensure stable operation.
- All wiring connections were double-checked before powering up the circuit to prevent damage or incorrect operation.

Programming Logic

- The purpose of the Arduino code was to control two DC motors using the L298P Motor Driver Shield.
- Each motor's speed was controlled by a PWM pin, while direction was managed through two digital pins.
- All motor control pins were defined and set as output in the setup section of the program.
- In the main loop, a custom function named Motor() was repeatedly called to run Motor A, Motor B, or both motors together.
- The motors were programmed to rotate forward, reverse, and stop by changing direction signals and PWM values.
- The Motor() function determined which motor to activate and in which direction, using switch and conditional statements.
- The analogWrite command adjusted the speed of each motor, allowing simultaneous or independent motor control.

Code Used

```
const int ENA = 10; // Control speed of motor A
const int ENB = 11; // Control speed of motor B
const int INA = 12; // Control direction of motor A
const int INB = 13; // Control direction of motor B

void setup()
{
  pinMode(ENA, OUTPUT);
  pinMode(INA, OUTPUT);
  pinMode(INB, OUTPUT);
  pinMode(ENB, OUTPUT);
}

void loop()
{
  Motor('A', 'F', 255); //Control motor A
  delay(5000);
  Motor('B', 'F', 255); //Control motor B
  delay(5000);
  Motor('C', 'F', 255); //Control both motor simultaneously
  delay(5000);
  Motor('C', 'B', 255);
  delay(5000);
}
```

```

    Motor('C', 'B', 0); //soft stoping both motor
simultaneously
    delay(5000);
}

void Motor(char mot, char direction, int
speed)
{
    switch (mot)
    {
        case 'A':
            if(direction == 'F') //Forward
            {
                digitalWrite(INA, HIGH);
            }
            else if(direction == 'B') //Backward
            {
                digitalWrite(INA, LOW);
            }
            analogWrite(ENA, speed);
            break;

        case 'B':
            if(direction == 'F')
            {
                digitalWrite(INB, HIGH);
            }
            else if(direction == 'B')
            {
                digitalWrite(INB, LOW);
            }
            analogWrite(ENB, speed);
            break;

        case 'C':
            if(direction == 'F')
            {
                digitalWrite(INA, HIGH);
                digitalWrite(INB, HIGH);
            }
            else if(direction == 'B')
            {
                digitalWrite(INA, LOW);
                digitalWrite(INB, LOW);
            }
            analogWrite(ENA, speed);
            analogWrite(ENB, speed);
            break;
    }
}

```

Control Algorithm

1. Pin Setup

- Motor A uses ENA (D10) for speed and INA (D12) for direction.
- Motor B uses ENB (D11) for speed and INB (D13) for direction.
- All pins are set as output to control the L298P Motor Driver Shield.

2. Variable Declaration

- Four integer constants (ENA, ENB, INA, INB) store the assigned pin numbers.

- A custom function Motor(mot, direction, speed) controls one or both motors based on input parameters.

3. Setup Function

- All control pins are initialized as OUTPUT using pinMode().
- This prepares the Arduino to send speed and direction signals to both motors.

4. Main Loop

- The program runs five control states in sequence:
- Motor A forward (Motor('A','F',255))
- Motor B forward (Motor('B','F',255))
- Both forward (Motor('C','F',255))
- Both backward (Motor('C','B',255))
- Stop both (Motor('C','B',0))

5. Motor Function

- The Motor() function selects the target motor and direction using a switch statement.
- Forward sets the direction pin HIGH, backward sets it LOW.
- Speed is adjusted using analogWrite() for PWM control on each motor.

5.0 DATA COLLECTION

Observation table:

PWM Value	Direction	Speed (RPM)
255	Forward	1113
128	Forward	220
255	Reverse	1150
64	Reverse	83

Coding: (See appendix A)

The coding using arduino IDE was very simple in terms of allowing the motor speed and direction to be HIGH. When the INA and INB were HIGH the direction turned clockwise and counter-clockwise when it was LOW.

6.0 DATA ANALYSIS

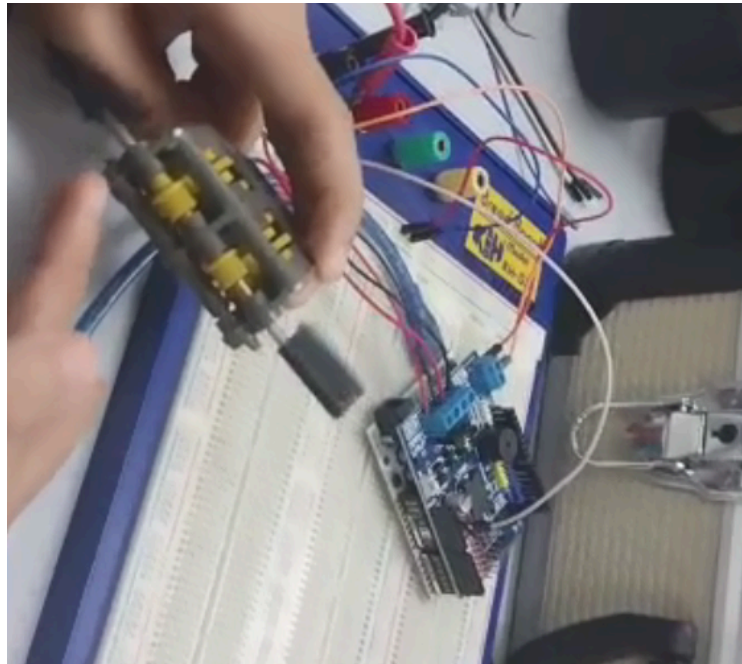
Based on the observation table;

- At PWM 255 (Forward Direction), the motor rotates slowly at first but continues to reach max speed after 1 sec of rotation.
- At PWM 128 (Forward Direction), the motor rotates smoothly with moderate noise generated.

- At PWM 255 (Reverse Direction), the motor rotates counter-clockwise the exact speed as forward direction for the same PWM.
- At PWM 64 (Reverse Direction), the motor rotates slowly counter-clockwise

A simple observation from the data collected shows that the RPM will increase linearly with the PWM value. A higher PWM, makes a higher duty cycle, that allows higher average voltage that results in faster rotation and vice versa.

7.0 RESULTS



The experiment successfully demonstrated the control of a DC motor using the L298P Motor Driver Shield and Arduino UNO. The system effectively responded to code commands written and uploaded through the Arduino IDE, allowing precise adjustment of both motor speed and direction. The Arduino code was simple and straightforward, using GPIO pins (IN1 and IN2) to control each motor's direction and PWM signals (ENA and ENB) to vary their speeds. The experiment verified that the IN1 and IN2 pins correctly controlled the direction of their respective motors. When both INA and INB pins were set to HIGH, the motor rotated in a clockwise direction, while setting them LOW caused the motor to rotate counterclockwise. Adjusting the PWM duty cycle on the ENA pin successfully changed the motor's speed, higher PWM values resulted in faster rotation, while lower values produced slower motion.

The motor reacted smoothly to code changes, confirming stable communication between the Arduino and the L298P shield. The overall system integration between hardware (Arduino UNO, L298P shield, DC motor) and software (Arduino IDE code) functioned reliably. The motor's direction changes were immediate and consistent, and the PWM control provided predictable speed variation. Overall, the results verified that the L298P Motor Driver Shield can effectively drive a DC motor with clear and responsive control over both speed and direction. The experiment successfully demonstrated the principles of PWM speed control and GPIO-based directional control using simple Arduino programming.

TASK

1. Explain the function of the ENA and ENB pins.

ENA and ENB are the enable pins for Motor A and Motor B. When connected to a PWM output, they control the speed of each motor by modulating the enable input.

If ENA/ENB = HIGH, the motor is enabled and will operate according to the PWM signal.

If ENA/ENB = LOW, the motor is disabled and will not run, regardless of the direction inputs.

2. Describe the reason PWM is used for speed control.

PWM is used because it provides high efficiency in switching power on and off, resulting in low power loss. A higher duty cycle gives a higher average voltage, making the motor run faster. A lower duty cycle gives a lower average voltage, causing the motor to run slower. This allows accurate and efficient speed control.

3. Describe the outcome when both IN1 and IN2 are set HIGH.

When IN1 and IN2 are both HIGH, it means that the motor will spin in the same direction (clockwise). On the L298P driver, setting both direction pins HIGH forces the H-bridge into a configuration that drives the motor forward in a fixed direction. This ensures consistent clockwise rotation as long as both inputs remain HIGH.

4. Explain how braking can be implemented using the L298P.

Braking can be implemented by allowing the speed to be 0 in the coding. When the PWM value is set to 0 on the ENA and ENB pin, the motor receives no power, causing it to stop. This acts as a simple braking method because reducing the speed to zero cuts off energy to the motor and slows it down until it stops.

5. Modify the code to control two DC motors simultaneously.

Refer **Appendix A: Coding**. The coding already shows the implementation for both motors moving simultaneously. The program sends the same direction and PWM signals to ENA/ENB and INA/INB, allowing both Motor A and Motor B to operate together at the same speed and direction.

8.0 DISCUSSION

This experiment's primary goal was to test how effectively a DC motor could be controlled using an Arduino Uno and an L298P Motor Driver Shield. From the results, it was observed that the motor responded correctly to changes in direction and speed. When the direction pins (IN1 and IN2) were set to opposite states (HIGH/LOW or LOW/HIGH), the motor's rotation successfully reversed. This confirmed that the direction control setup between the Arduino and the motor driver worked as intended.

The experiment also proved that Pulse Width Modulation (PWM) is a reliable method for speed control. When the PWM value sent to the ENA pin was high (such as 255), the motor ran at full speed, representing a 100% duty cycle where power was continuously supplied. In contrast, when lower PWM values like 64 or 128 were applied, the motor rotated more slowly because the duty cycle was reduced, meaning the average voltage supplied was lower. This shows that the L298P Motor Driver Shield, together with the Arduino, provides an effective and low-cost solution for controlling motor movement in projects such as robotics and automated systems.

Sources of Error and Limitations:

1. Motor Malfunctions

- During testing, both motors did not rotate simultaneously as expected. One motor started later or rotated slower than the other. This issue was likely caused by differences in motor conditions, uneven wiring, or unbalanced power delivery, leading to inconsistent rotation.

2. Estimation Error

- The recorded motor speed (in RPM) was only estimated by observation instead of being measured with a tachometer. This made the results less accurate and may have introduced human error during data collection.

3. Power Supply Fluctuations

- Slight changes in the external power supply voltage may have affected the motor's maximum speed and performance at certain PWM values. This could cause small variations in the observed motor speed.

Improvements:

1. Check and tighten all wiring connections to make sure both motors receive balanced power and rotate at the same time.
2. Use a tachometer or speed sensor to obtain more accurate motor speed readings instead of relying on estimation.
3. Ensure a stable and regulated power supply to avoid voltage drops that can affect motor performance.

9.0 CONCLUSION

In conclusion, this experiment successfully demonstrated the control of a DC motor using the L298P Motor Driver Shield and Arduino UNO. The system effectively integrated hardware and software to regulate both the speed and direction of the motor through Pulse Width Modulation (PWM) and GPIO control signals. The Arduino IDE allowed simple and efficient programming, enabling smooth changes in motor rotation and speed. Through this experiment, we learned that IN1 and IN2 pins play a crucial role in controlling the direction of the motors. IN1 manages the direction of Motor A, while IN2 controls Motor B. When both pins were set to HIGH, the motor rotated clockwise, and when set to LOW, it rotated counterclockwise. Adjusting the PWM duty cycle at the ENA and ENB pins directly affected the motor's rotational speed, where higher PWM values resulted in faster motion. Although minor inconsistencies such as uneven rotation and slight power fluctuations were observed, the overall system performed reliably and achieved all experimental objectives. This experiment strengthened our understanding of motor driver interfacing, PWM-based control, and directional logic, which are essential concepts in mechatronics and robotics applications. Overall, the project successfully demonstrated how the L298P Motor Driver Shield can provide a practical and efficient method for controlling dual DC motors with precise speed and direction management.

10.0 RECOMMENDATIONS

For improved performance, it is recommended to use a tachometer or speed sensor to accurately measure the motor's RPM instead of relying on visual estimation. This would provide more precise data for analyzing the relationship between PWM values and motor speed. The system's reliability can also be enhanced by using a stable and regulated external power supply to ensure consistent motor operation and prevent voltage drops that may affect performance.

To achieve better synchronization when controlling two motors simultaneously, the use of matched DC motors with equal ratings is advised. Adding filter capacitors across the motor terminals can help reduce electrical noise and protect the driver circuit from voltage spikes. For more advanced control, future improvements could include implementing feedback control using sensors (such as encoders) to maintain constant speed under varying loads, or upgrading to a motor driver with higher current capacity for improved efficiency.

Integrating these recommendations would lead to a more stable, accurate, and versatile DC motor control system suitable for more complex mechatronic and robotic applications.

11.0 REFERENCES

Cirkit Designer. (n.d.). How to Use L298P drive shield: Examples, Pinouts, and Specs.

<https://docs.cirkitdesigner.com/component/a25f6e70-294e-42fd-b77c-9e9349b76b9a/l298p-drive-shield>

Cytron Technologies. (n.d.). Shield-L298P Motor Driver with GPIO.

<https://my.cytron.io/p-shield-l298p-motor-driver-with-gpio>

CytronTechnologies. (2016). L298P_MotorDriverShield [Software repository]. GitHub.

https://github.com/CytronTechnologies/L298P_MotorDriverShield

12.0 APPENDICES

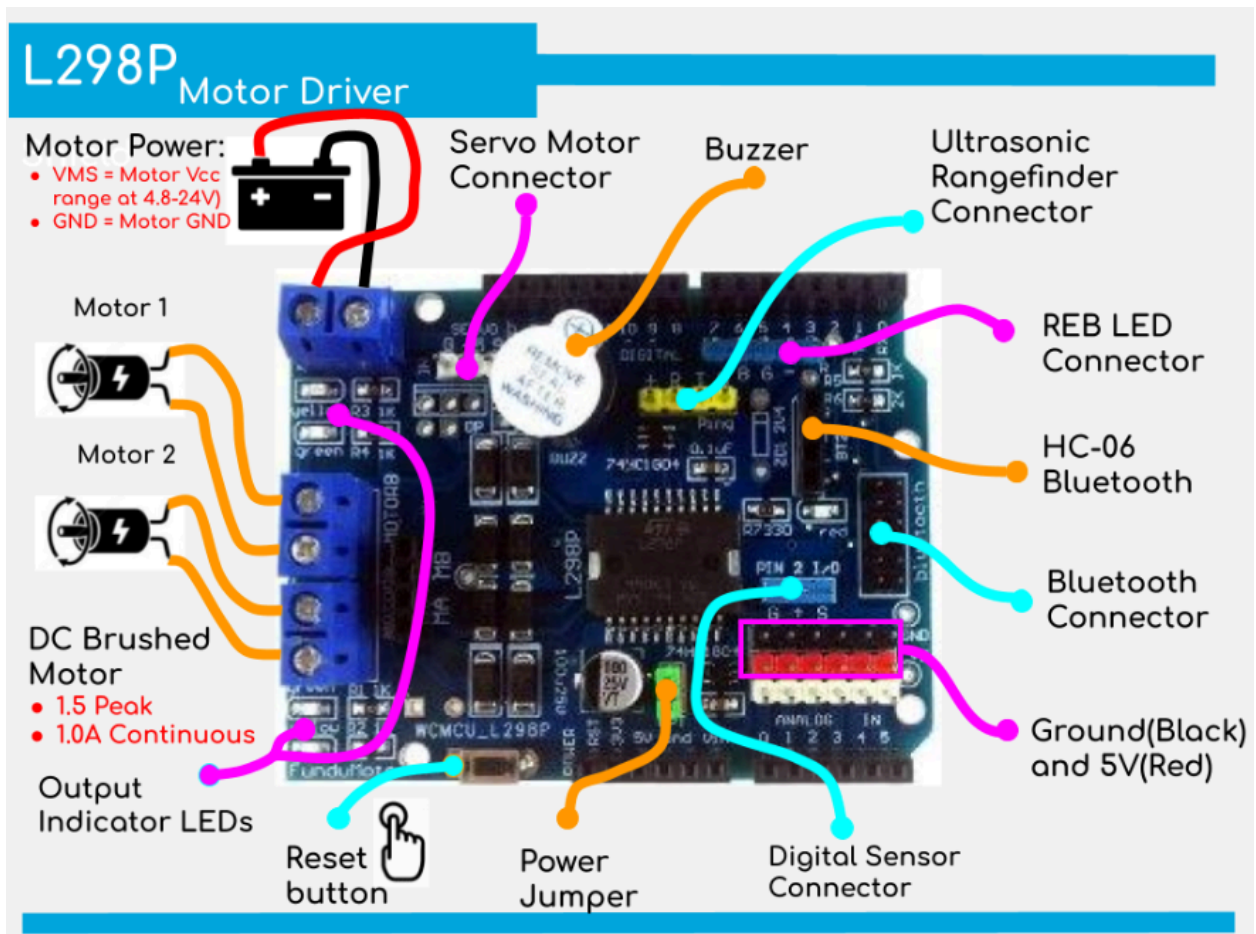
Appendix A: Coding

```
1  const int ENA = 10; // Control speed of motor A
2  const int ENB = 11; // Control speed of motor B
3  const int INA = 12; // Control direction of motor A
4  const int INB = 13; // Control direction of motor B
5
6  void setup()
7  {
8      pinMode(ENA, OUTPUT);
9      pinMode(INA, OUTPUT);
10     pinMode(INB, OUTPUT);
11     pinMode(ENB, OUTPUT);
12 }
13
14 void loop()
15 {
16     Motor('A', 'F', 255); //Control motor A
17     delay(5000);
18     Motor('B', 'F', 255); //Control motor B
19     delay(5000);
20     Motor('C', 'F', 255); //Control both motor simultaneously
21     delay(5000);
22     Motor('C', 'B', 255);
23     delay(5000);
24     Motor('C', 'B', 0); //soft stoping both motor simultaneously
25     delay(5000);
26 }
27
28 void Motor(char mot, char direction, int speed)
29 {
```

```
30     switch (mot)
31     {
32         case 'A':
33             if(direction == 'F') //Forward
34             {
35                 digitalWrite(INA, HIGH);
36             }
37             else if(direction == 'B') //Backward
38             {
39                 digitalWrite(INA, LOW);
40             }
41             analogWrite(ENA, speed);
42             break;
43
44         case 'B':
45             if(direction == 'F')
46             {
47                 digitalWrite(INB, HIGH);
48             }
49             else if(direction == 'B')
50             {
51                 digitalWrite(INB, LOW);
52             }
53             analogWrite(ENB, speed);
54             break;
55
```

```
56     case 'C':
57         if(direction == 'F')
58         {
59             digitalWrite(INA, HIGH);
60             digitalWrite(INB, HIGH);
61         }
62         else if(direction == 'B')
63         {
64             digitalWrite(INA, LOW);
65             digitalWrite(INB, LOW);
66         }
67         analogWrite(ENA, speed);
68         analogWrite(ENB, speed);
69         break;
70     }
71 }
```

Appendix B: Pin Layout from Cytron



13.0 STUDENT'S DECLARATION

Certificate of Originality and Authenticity

This is to certify that we are responsible for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or persons.

We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have **read** and **understand** the content of the total report and that no further improvement on the reports is needed from any of the individual contributors to the report.

Signature: 	Read	/
Name: AHMAD HARITH IMRAN BIN MOHD YUSOF	Understand	/
Matric No.: 2311581	Agree	/
Contribution : Assembler, table of content, results, conclusion, recommendation and student declaration.		

Signature: 	Read	/
Name: MOHAMMAD FARISH ISKANDAR BIN MOHD SYAHIDIN	Understand	/
Matric No.: 2311779	Agree	/
Contribution : Coder, experimental setup, data collection, data analysis, references and appendices.		

Signature: 	Read	/
Name: ARIFA AQILAH BINTI ABDUL HALIM	Understand	/
Matric No.: 2311530	Agree	/
Contribution : Assembler, abstract, introduction, material and equipment, methodology and discussion.		